

## Training Configuration and Summary:

Dataset used: rssi\_odom\_dataset\_12apr25\_ajust.parquet

model\_type: LSTM  
group\_sec: 1  
timesteps\_orig: 30  
timesteps: 30  
batch\_size: 256  
min\_units: 32  
max\_units: 256  
units\_step: 32  
dropout\_min: 0.2  
dropout\_max: 0.3  
dropout\_step: 0.1  
activation\_functions: ['tanh', 'relu', 'elu']  
optimizers: ['adam', 'rmsprop', 'Nadam']  
min\_layers: 1  
max\_layers: 2  
max\_epochs: 150  
hyperband\_iterations: 3  
forecast\_steps: 10

### LSTM Multi-Step Training Report

Report Generated: 2025-05-09 16:08:37

Training Start: 2025-05-09\_15-21-35

Training End: 2025-05-09\_15-41-56

Training Duration: 20.36 minutes

#### Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 1.0240

MSE: 1.6055

RMSE: 1.2671

R2 Score: 0.9115

#### Multi-Step Forecast:

Forecast Steps: 10

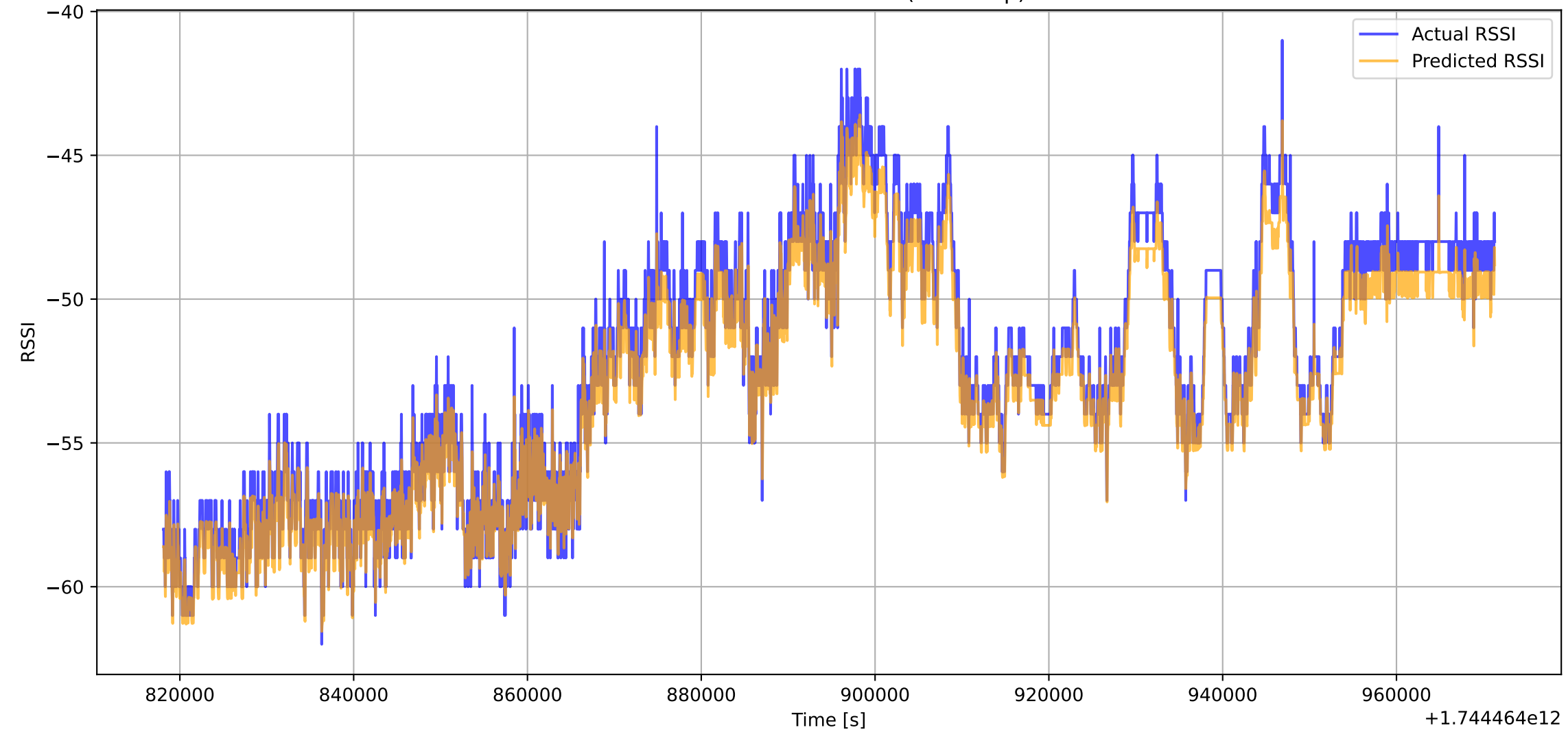
MAE: 0.0124

MSE: 0.0002

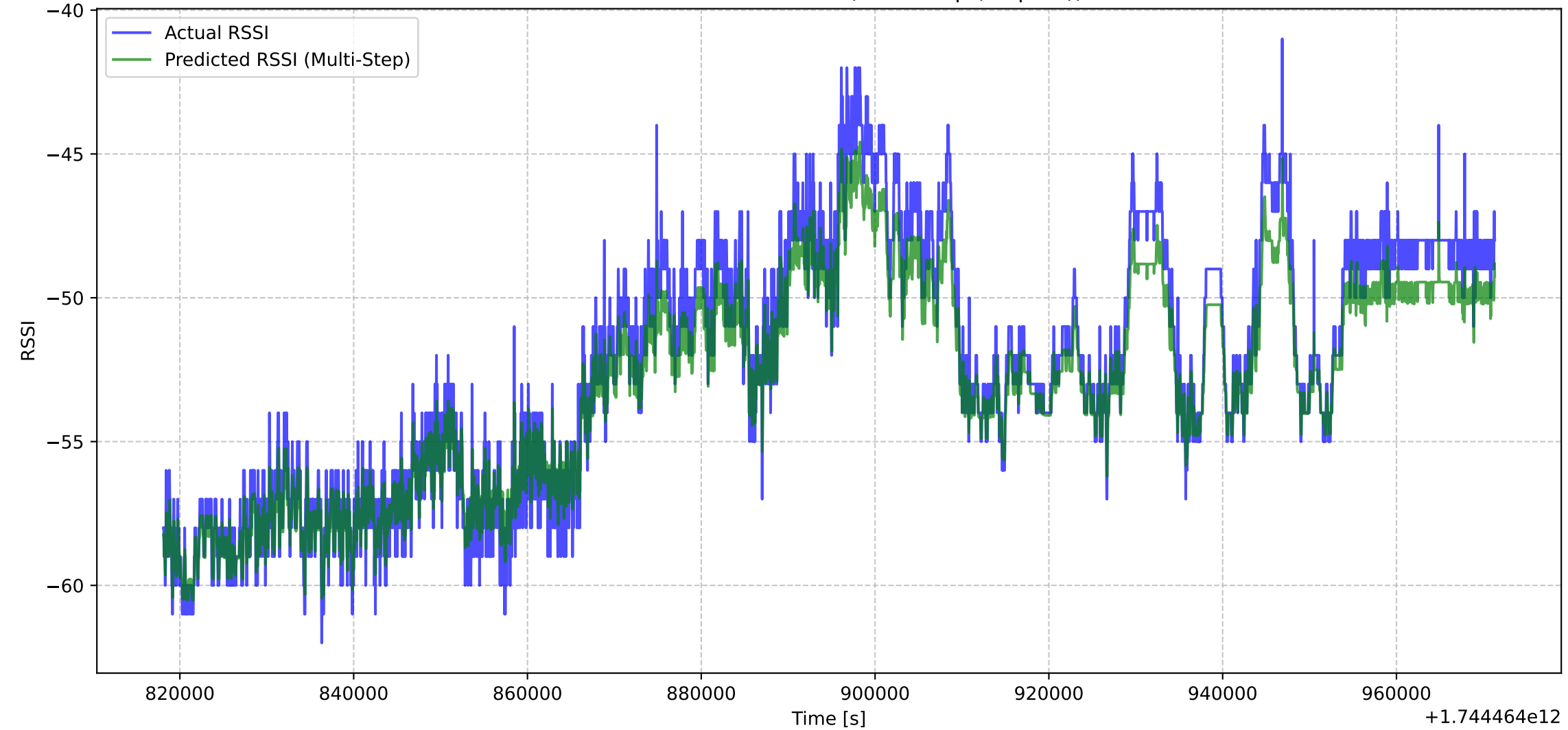
RMSE: 0.0148

R2 Score: 0.9461

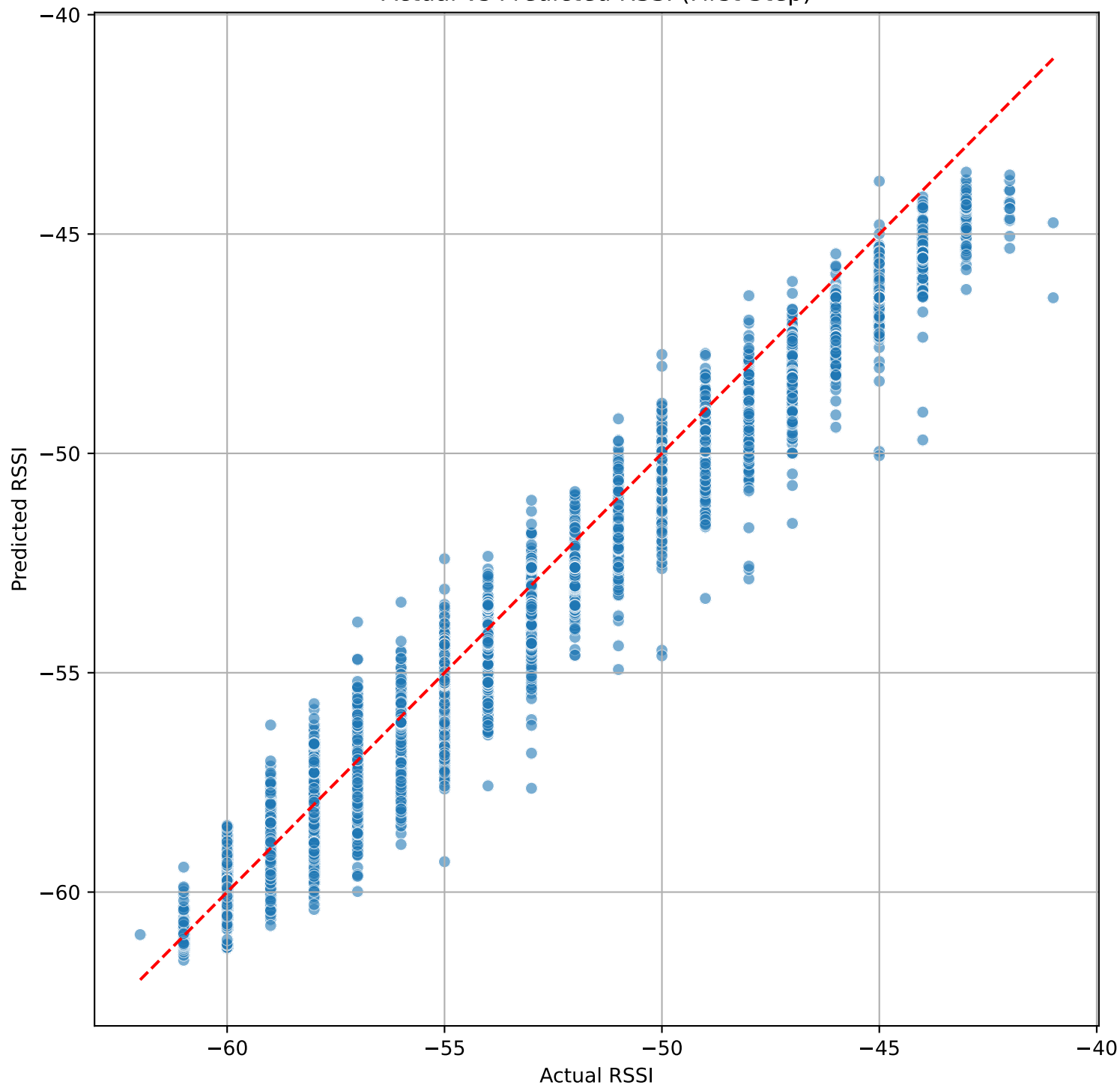
Actual vs Predicted RSSI (First Step)



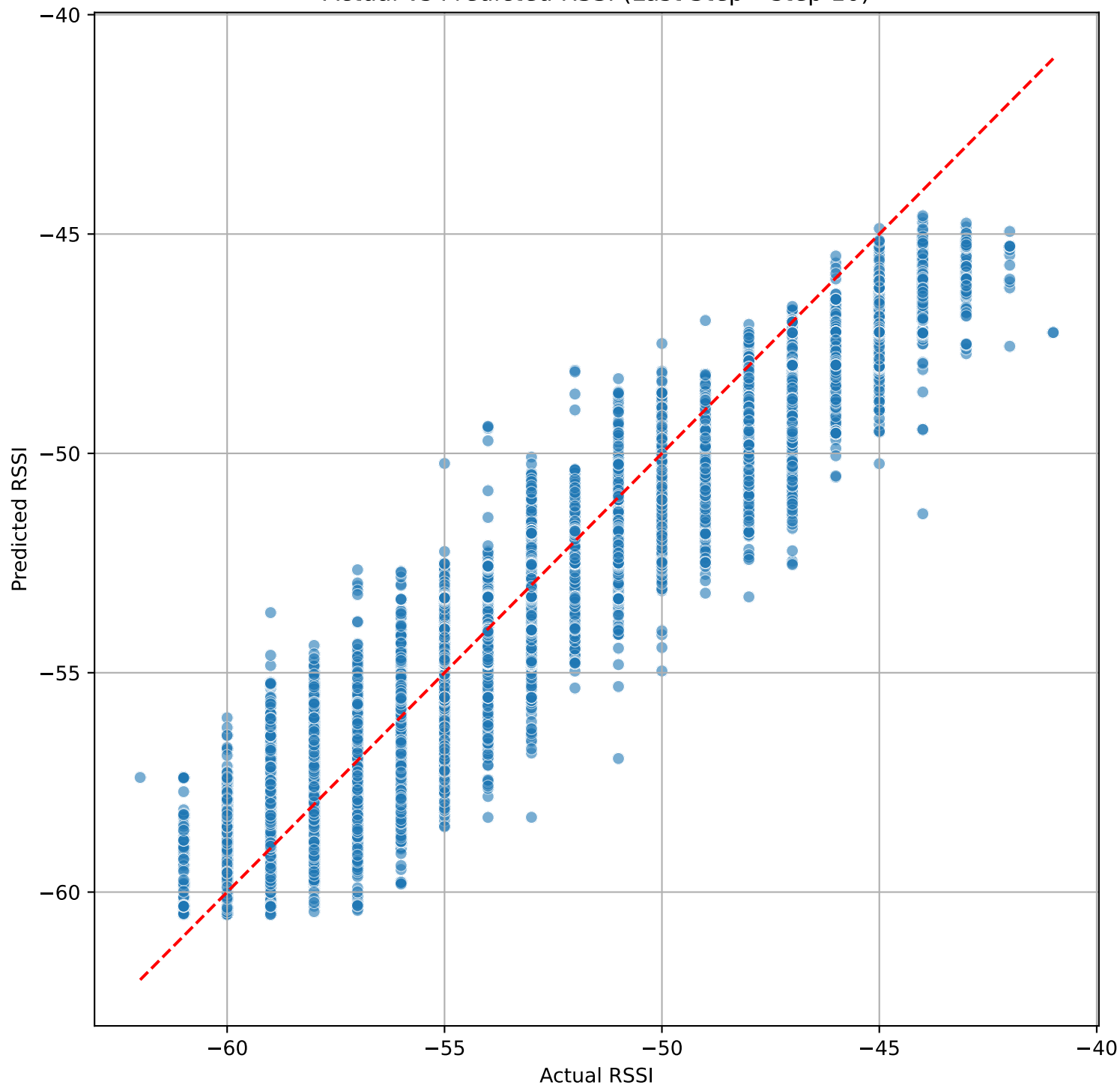
Actual vs Predicted RSSI (Multi-Step (step 10))



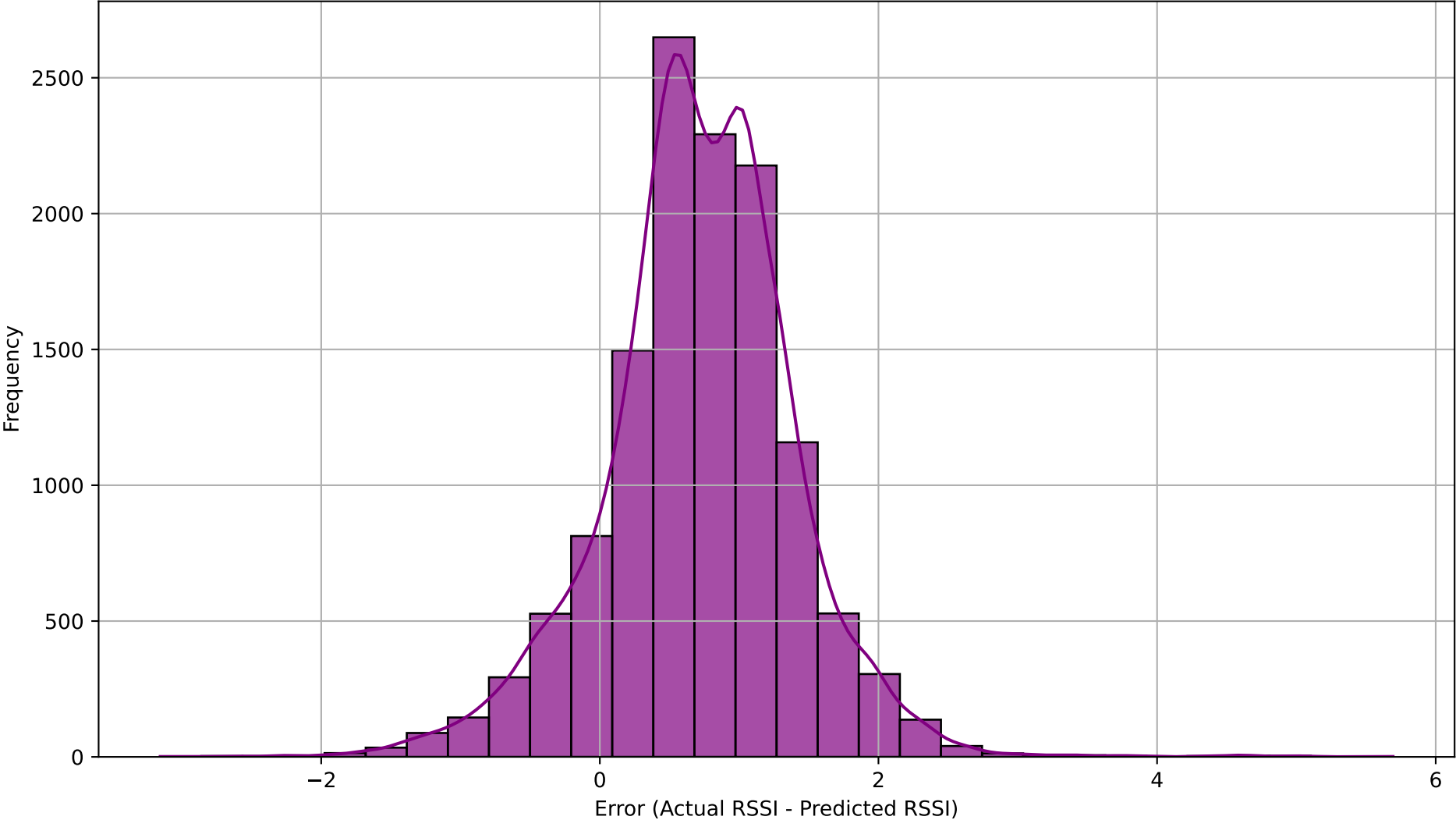
Actual vs Predicted RSSI (First Step)



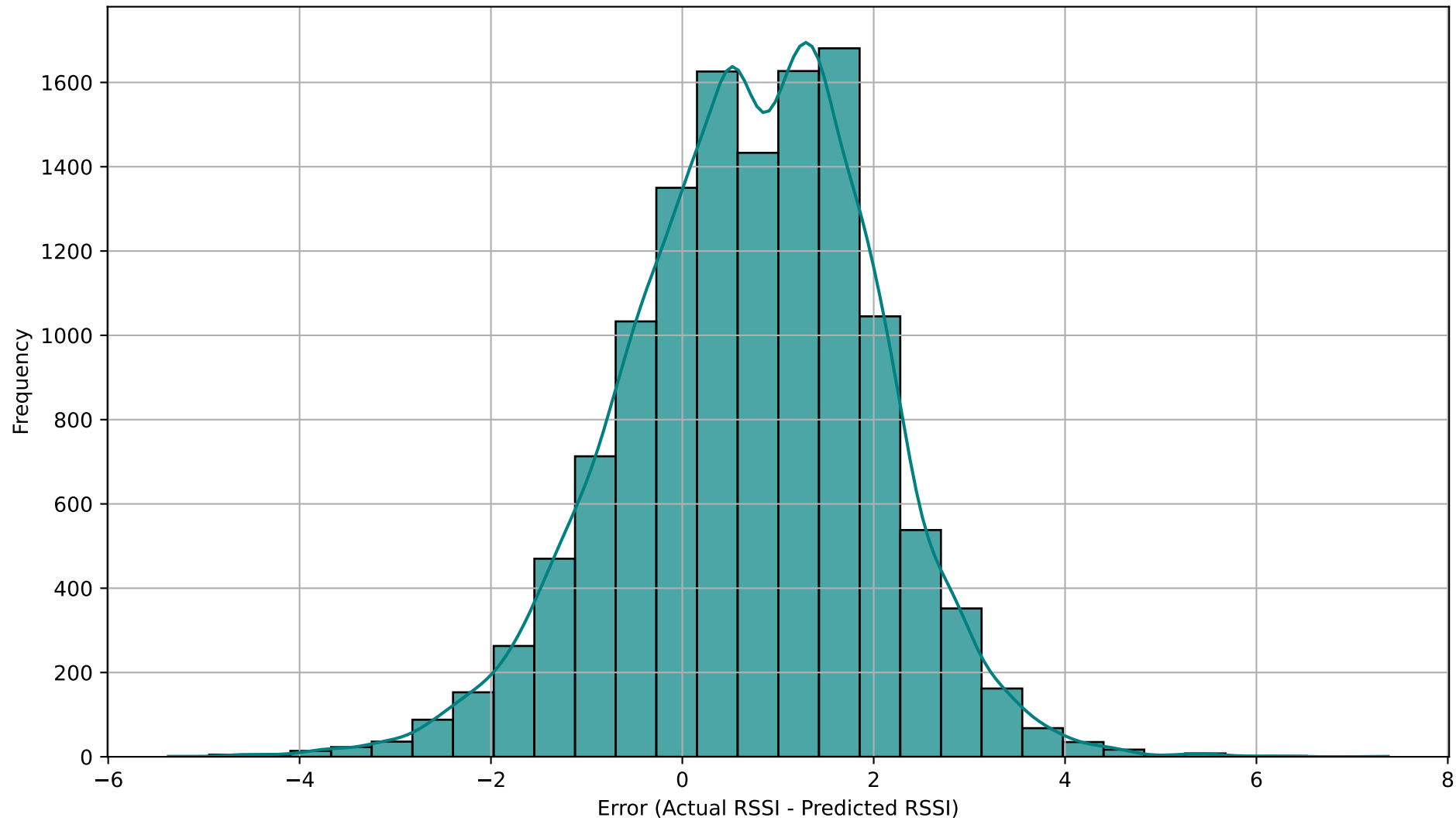
Actual vs Predicted RSSI (Last Step - step 10)



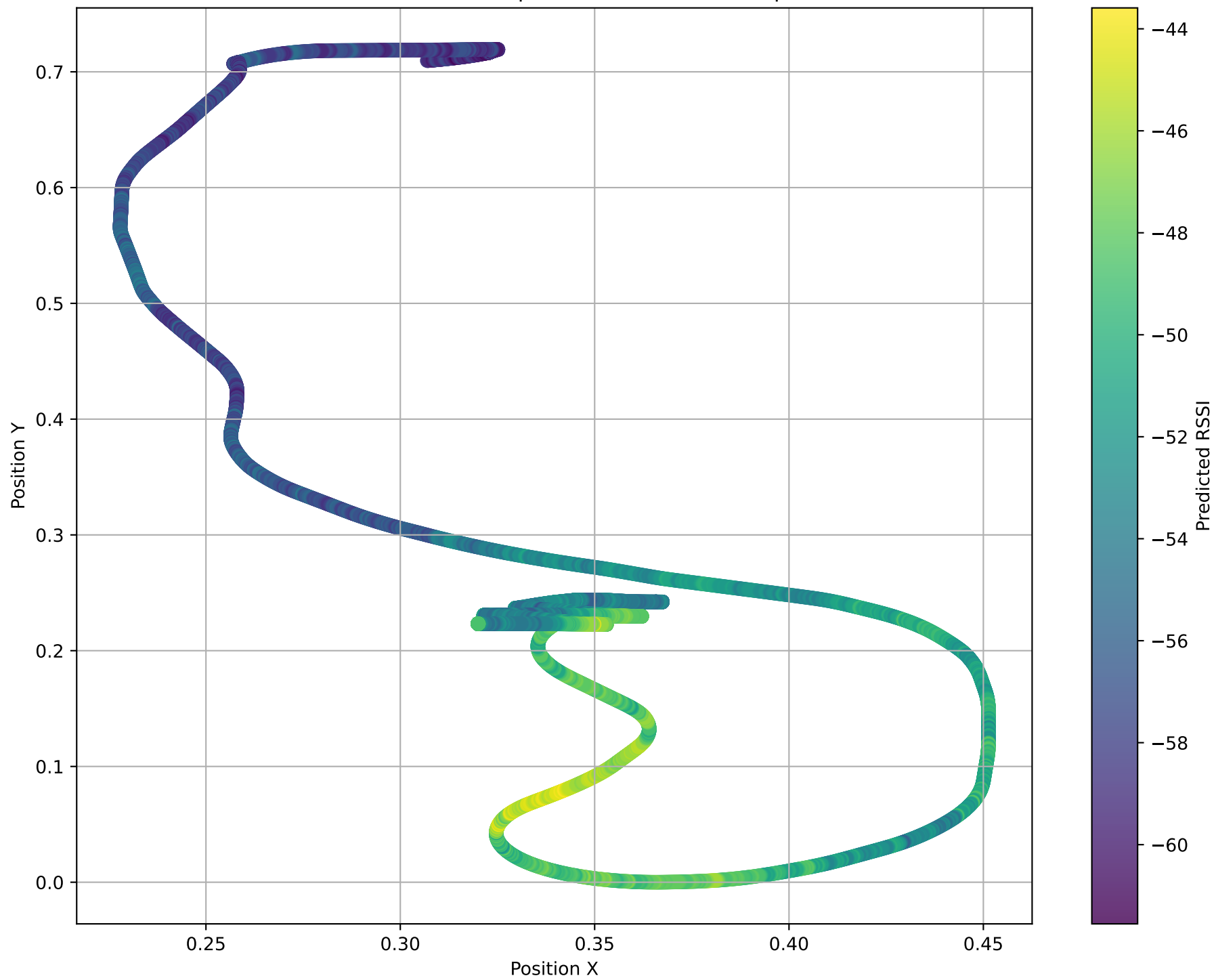
Distribution of Prediction Errors (First Step)



Distribution of Prediction Errors (Last Step - step 10)

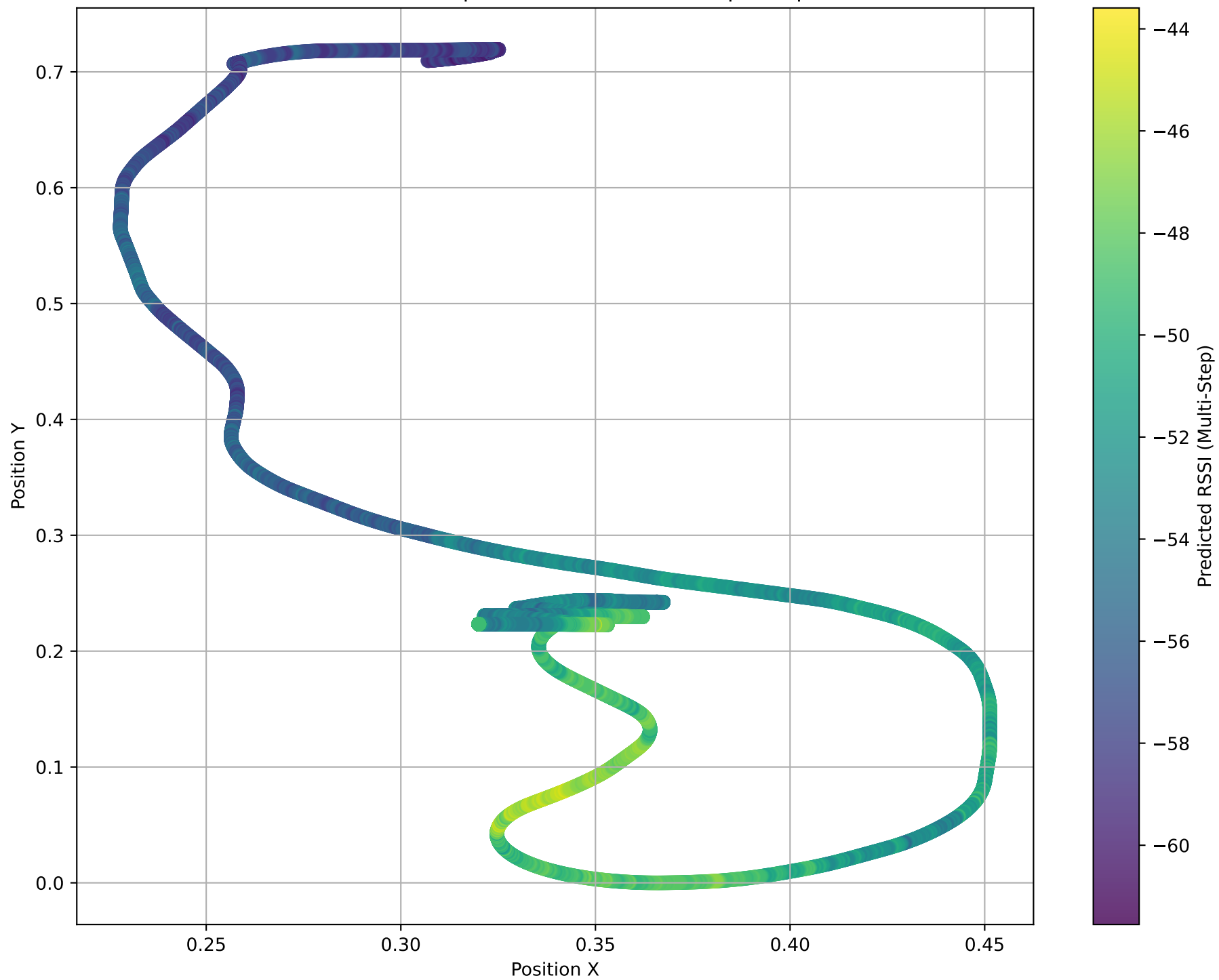


Predicted RSSI in Spatial Context (First Step)

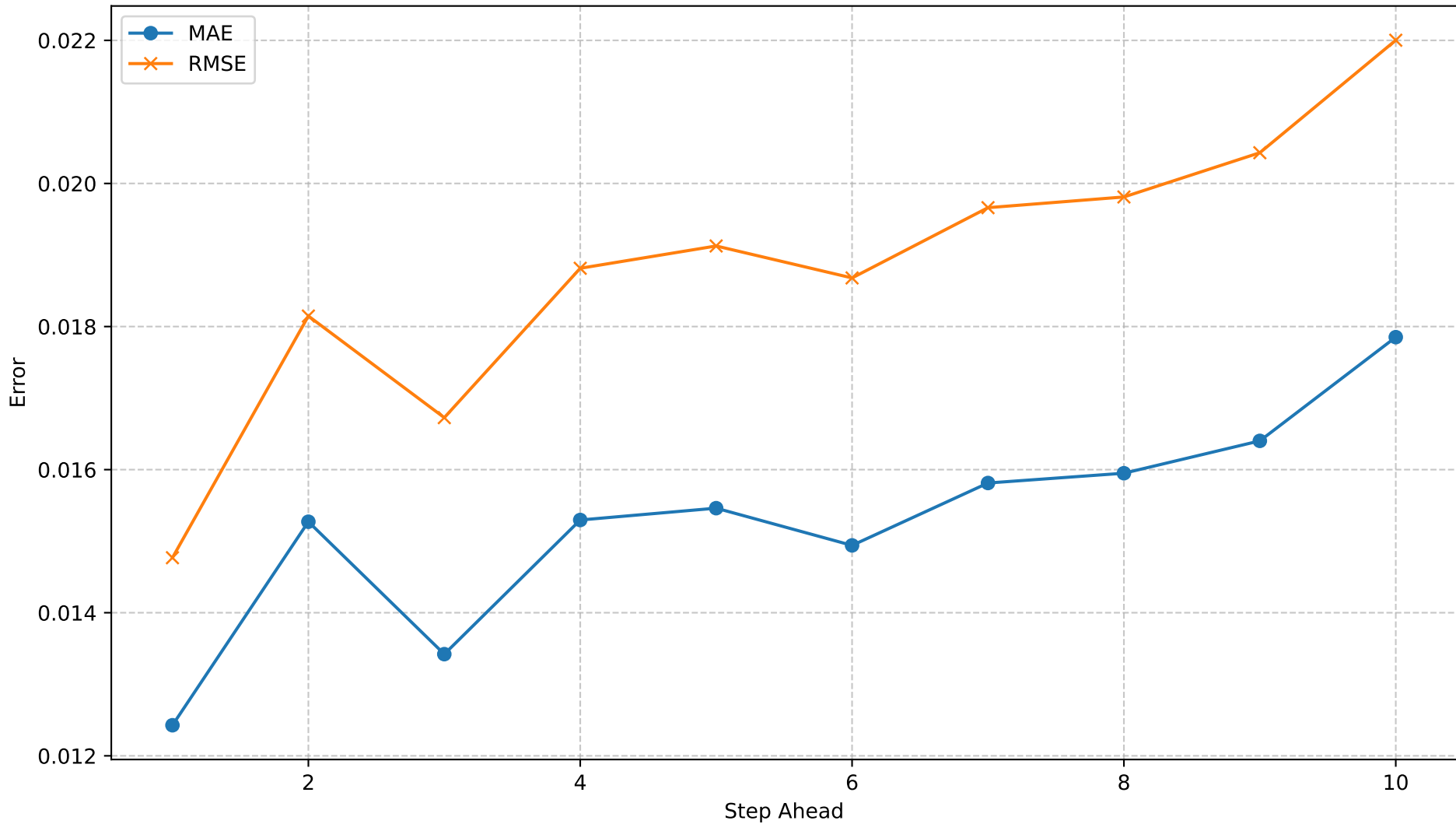




Predicted RSSI in Spatial Context (Multi-Step - step 10)



Multi-Step Forecast Error by Step



## Resumo Automático

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Dataset: rssi\_odom\_dataset\_12apr25\_ajust.parquet

Modelo: LSTM

Forecast steps: 10

MAE / RMSE (dB)

- 1º passo : 0.01 / 0.01
- Mediana : 0.02 / 0.02
- 10º passo : 0.02 / 0.02

Crescimento do erro: 43.7% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 10 (0.02), 9 (0.02), 8 (0.02)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.